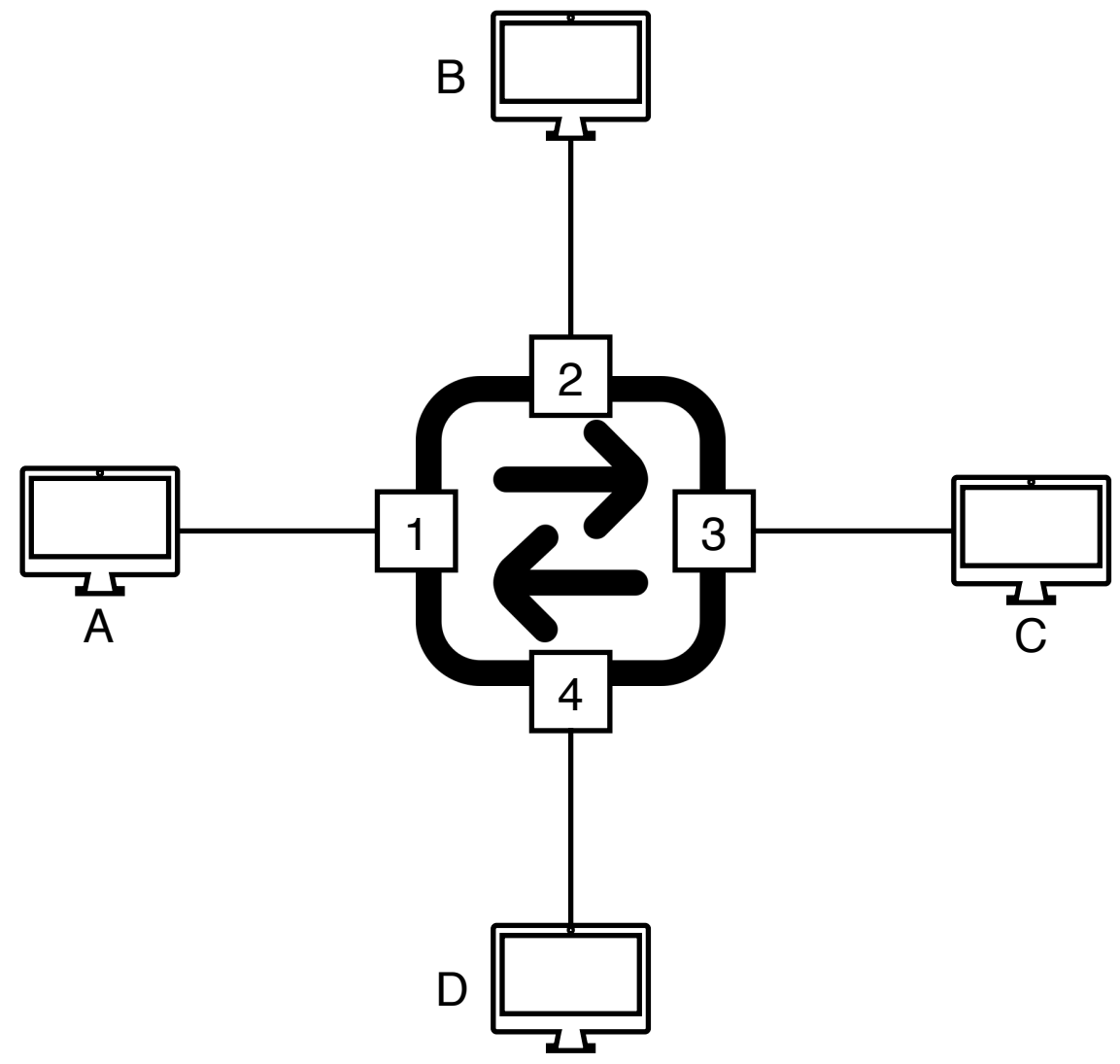
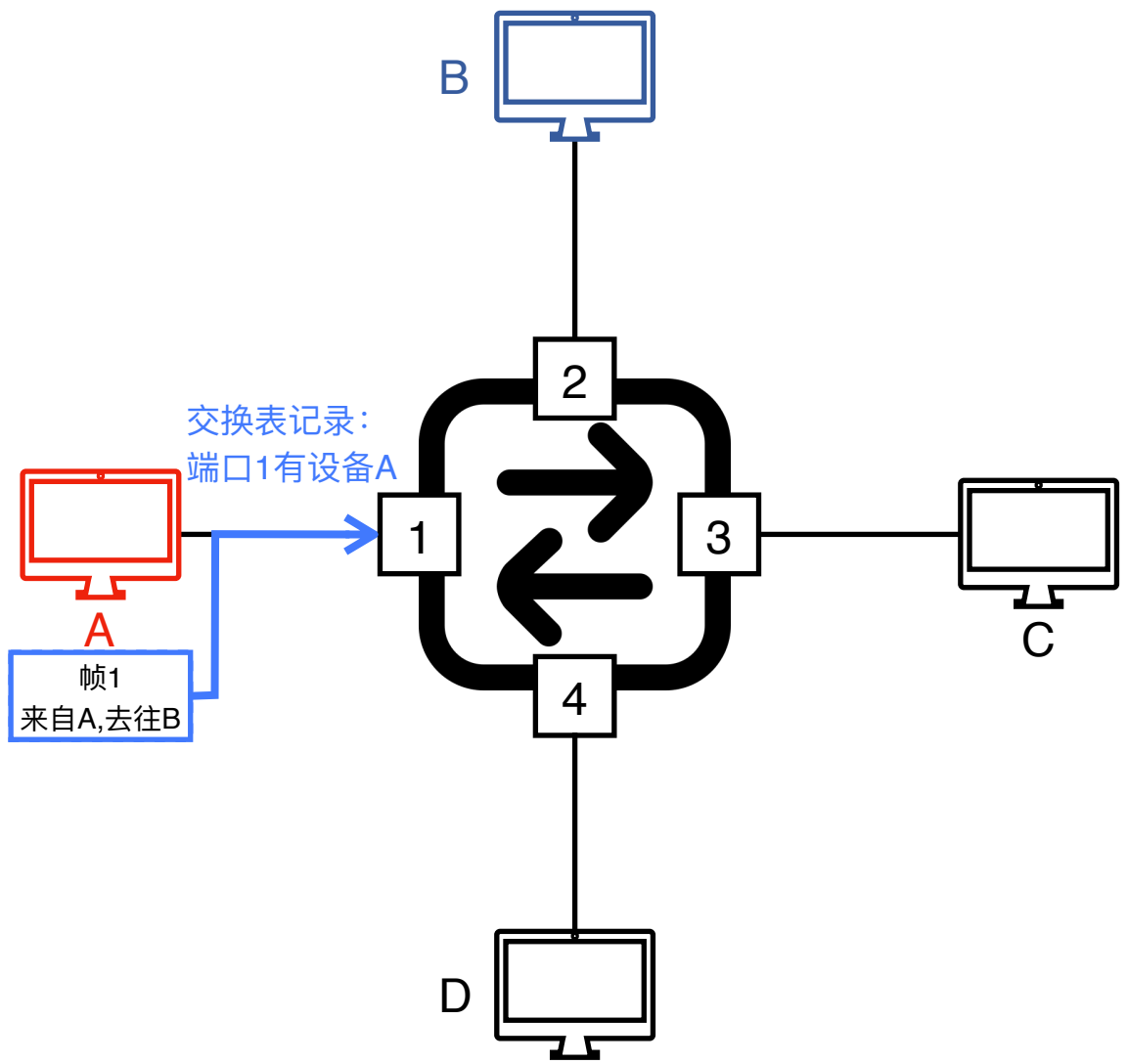




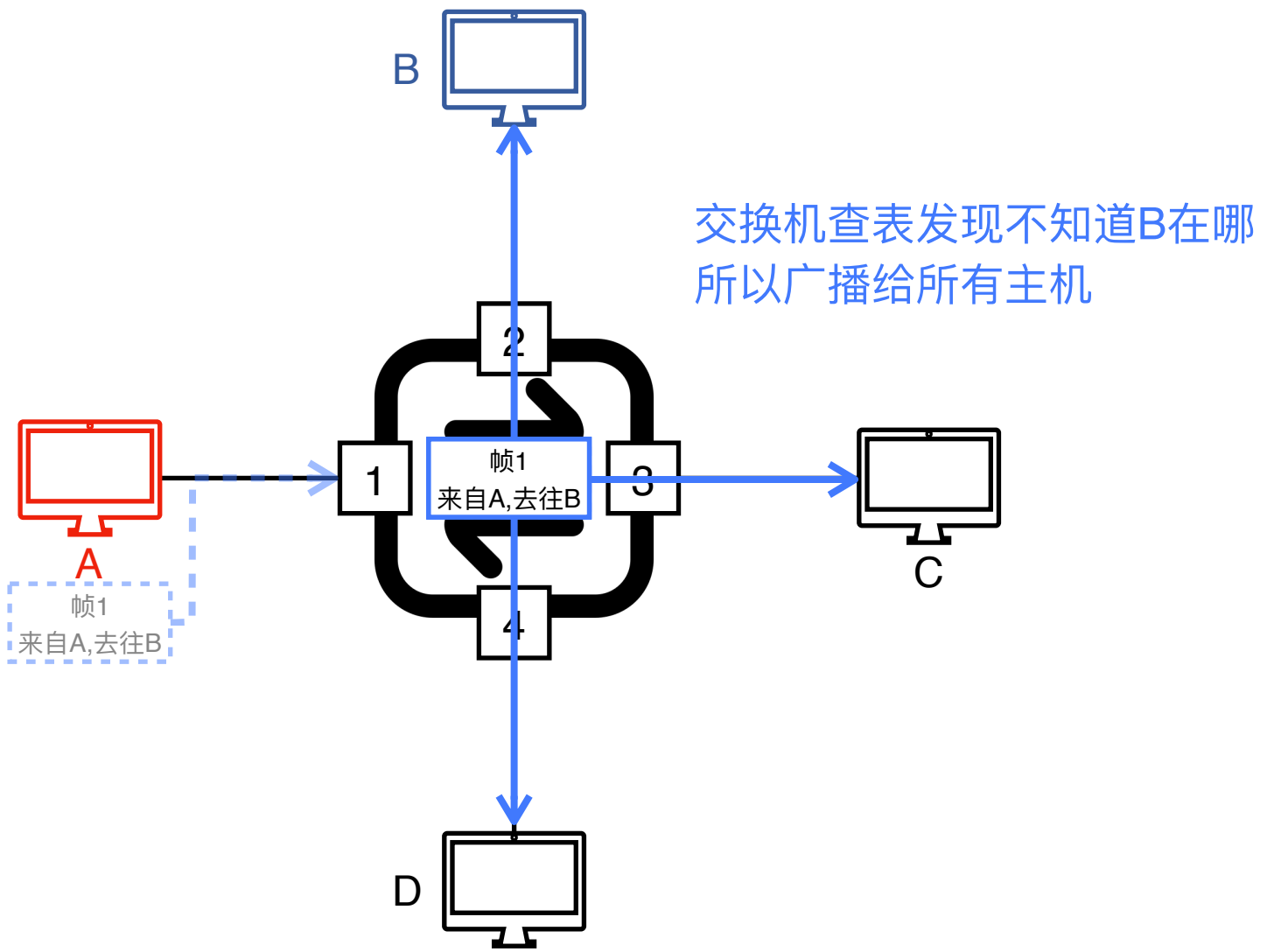
1.A->B

1	交换表		3
2	MAC地址	端口	4



1.A->B

1	交换表		3
2	MAC地址	端口	4
	A	1	



1.A->B

交换表	
MAC地址	端口
A	1



The diagram shows a switch with 4 ports. On the left, ports 1 and 2 are labeled in boxes. On the right, ports 3 and 4 are labeled in boxes. In the center is a switching table titled "交换表".

MAC地址	端口
A	1

B



帧1
来自A, 去往B

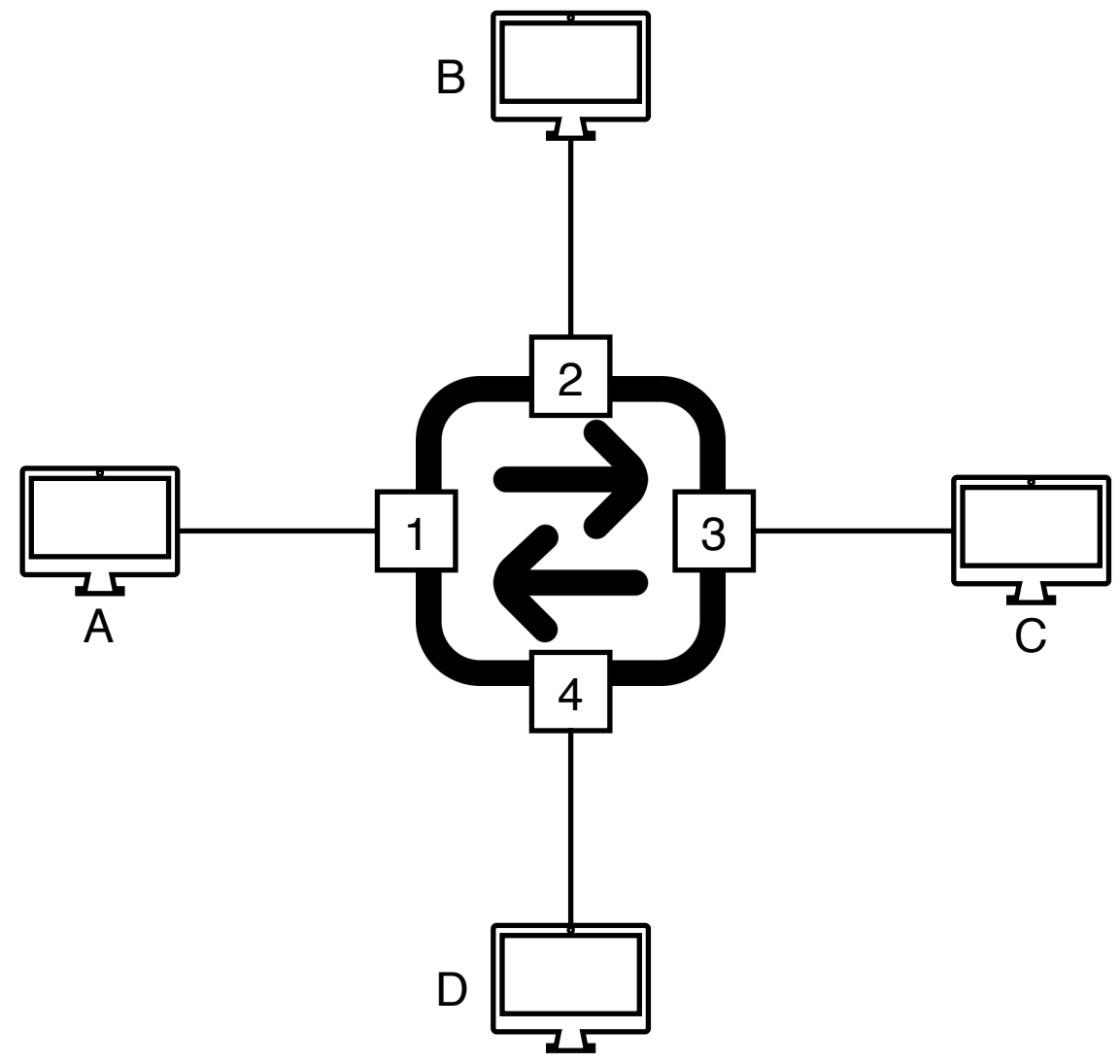
D 帧1
来自A,去往B

The diagram shows a central switch table with the title "交换表" (Switch Table). The table has two columns: "MAC地址" (MAC Address) and "端口" (Port). The first row contains the value "A" under "MAC地址" and "1" under "端口". The other three rows are empty. To the left of the table are two boxes labeled "1" and "2", and to the right are two boxes labeled "3" and "4", representing the four ports of the switch.

MAC地址	端口
A	1

The diagram shows a central switch table with the title "交换表" (Switch Table). The table has two columns: "MAC地址" (MAC Address) and "端口" (Port). The first row contains the value "A" under "MAC地址" and "1" under "端口". The other three rows are empty. To the left of the table are two boxes labeled "1" and "2", and to the right are two boxes labeled "3" and "4", representing the four ports of the switch.

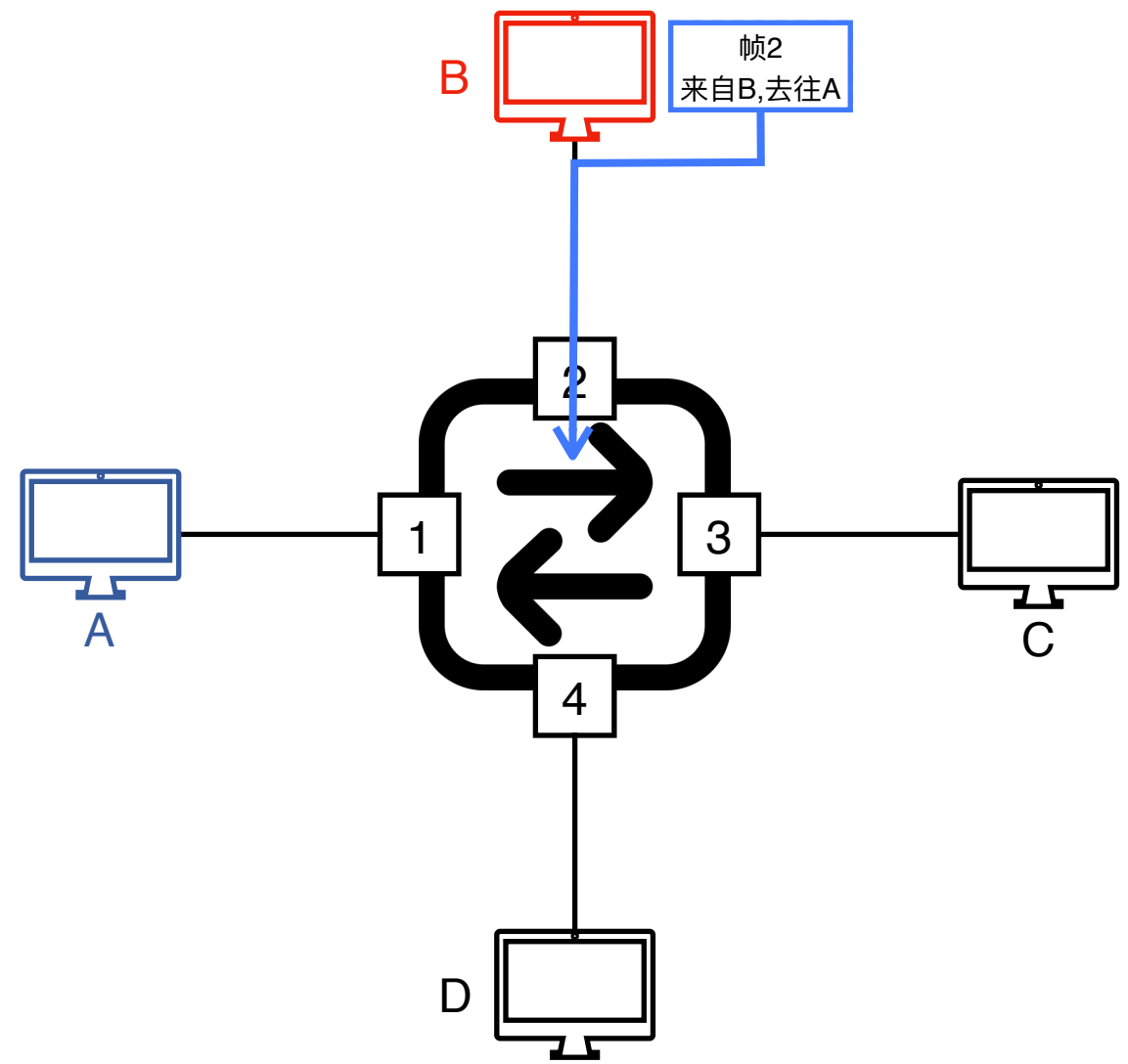
MAC地址	端口
A	1



1.A->B

1	交换表		3
2	MAC地址	端口	4
	A	1	

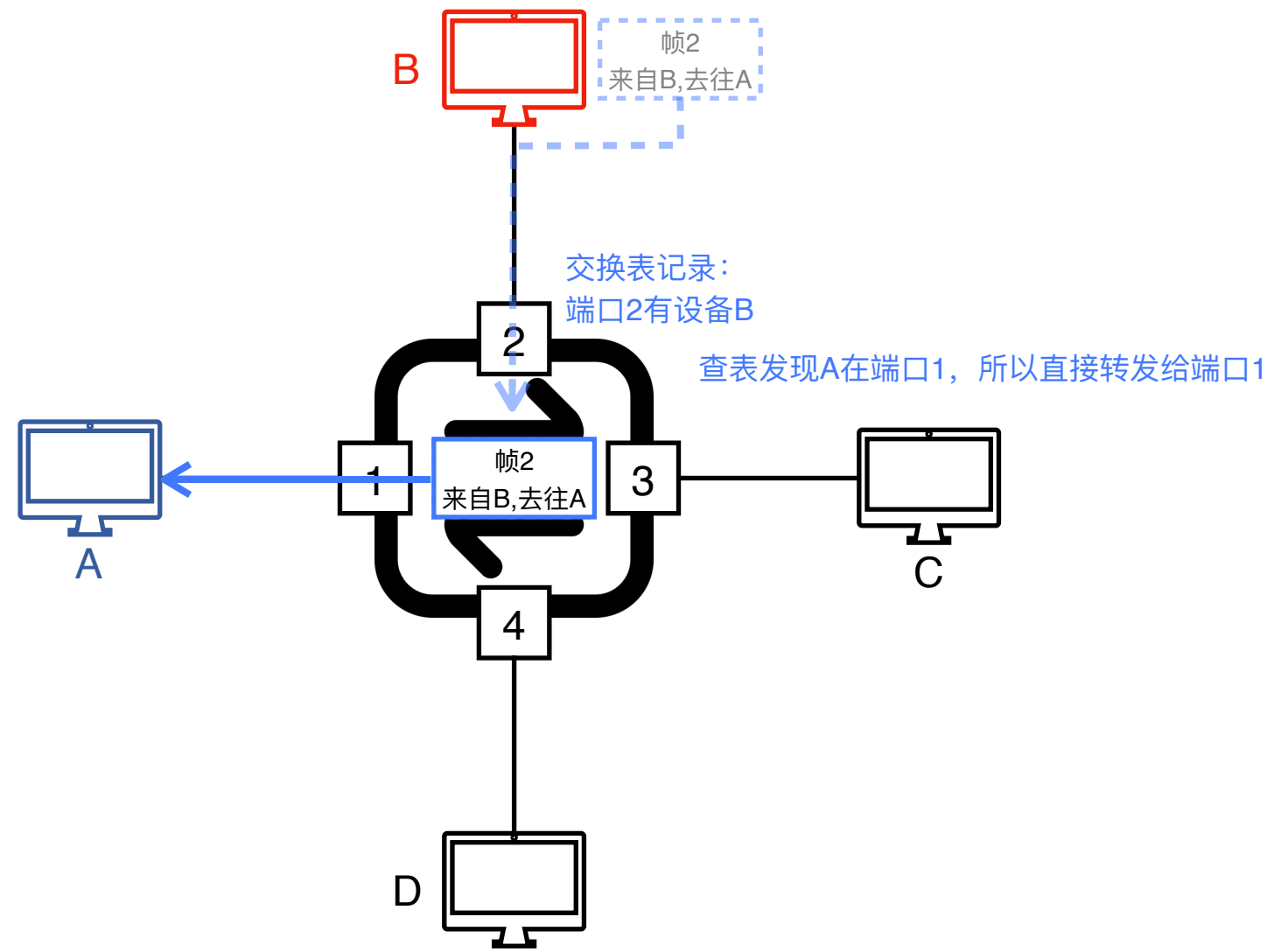
此时B再给A发送一帧



- 1.A->B
- 2.B->A

1	交换表		3
2	MAC地址	端口	4
	A	1	

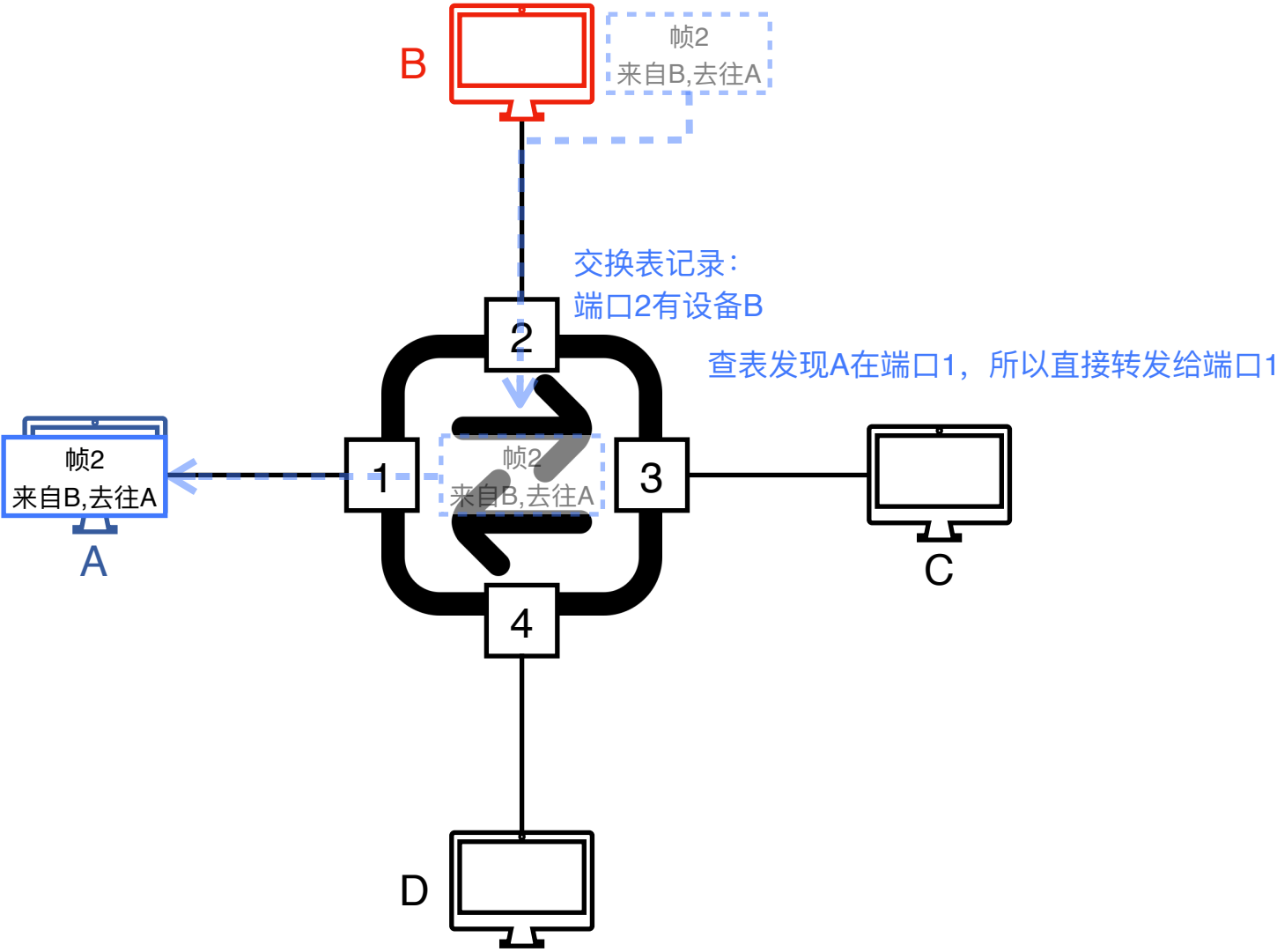
此时B再给A发送一帧



- 1.A->B
- 2.B->A

1	交换表		3
2	MAC地址	端口	4
	A	1	
	B	2	

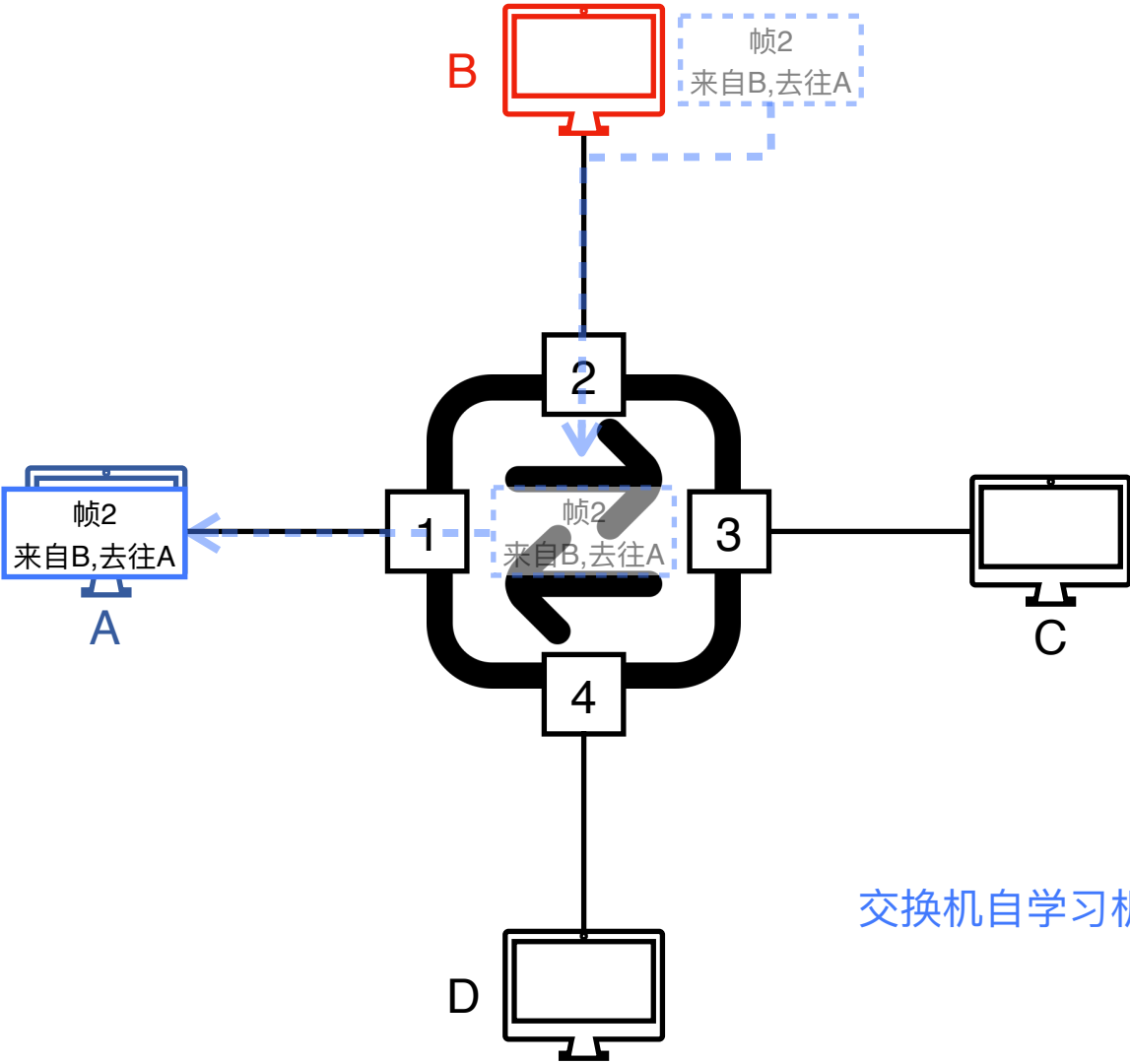
此时B再给A发送一帧



- 1.A->B
- 2.B->A

1	交换表		3
2	MAC地址	端口	4
	A	1	
	B	2	

此时B再给A发送一帧

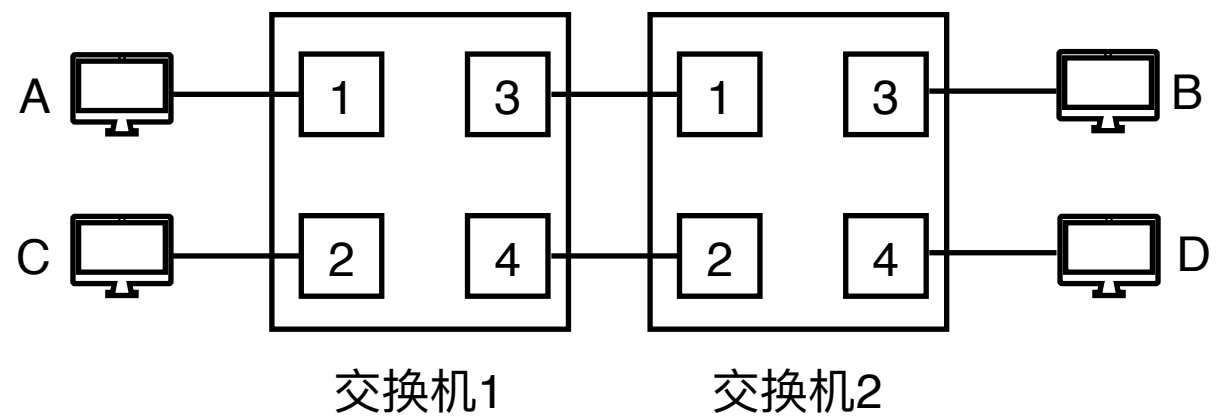


- 1.A->B
- 2.B->A

1	交换表		3
2	MAC地址	端口	4
	A	1	
	B	2	

交换机自学习机制：谁发送了帧，就记录谁的MAC和端口对应关系

兜圈问题



A向B发送一帧

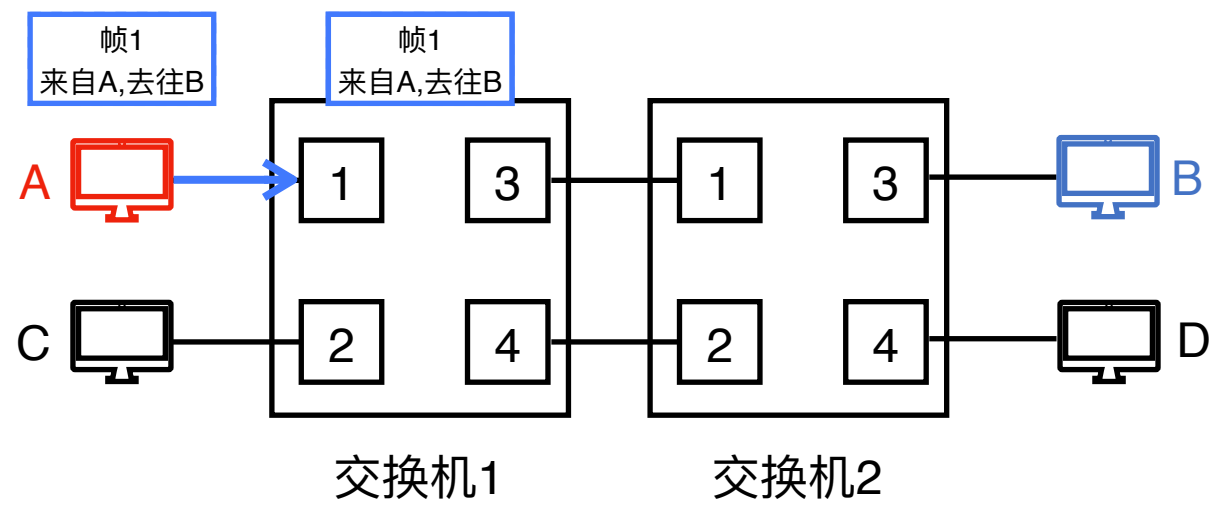
交换表1

MAC地址	端口

交换表2

MAC地址	端口

兜圈问题



A向B发送一帧

交换机1收到帧后，查表未果，进行广播,同时记录A在端口1

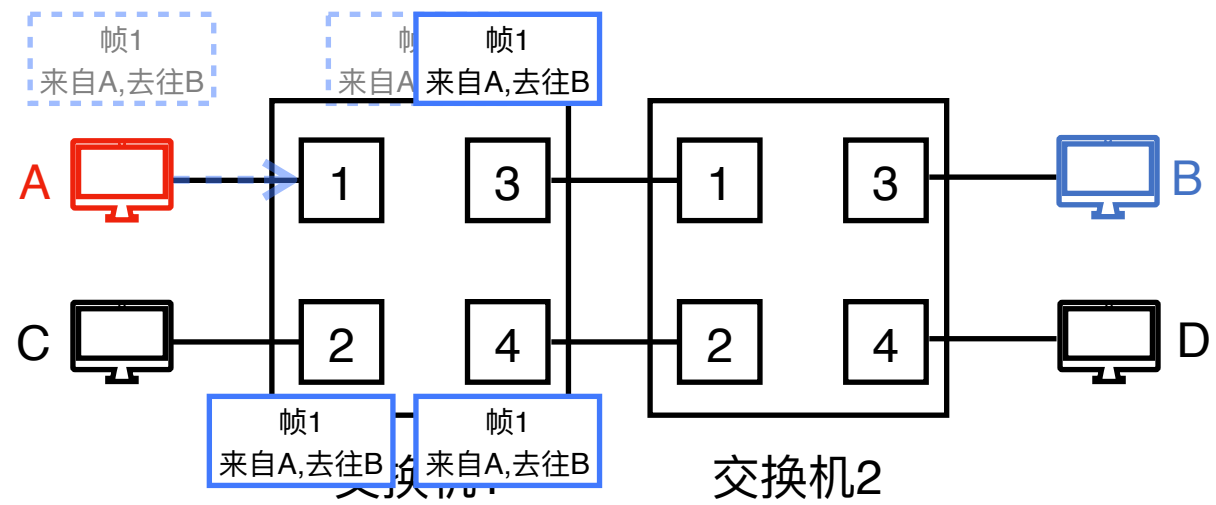
交换表1

MAC地址	端口
A	1

交换表2

MAC地址	端口

兜圈问题



A向B发送一帧

交换机1收到帧后，查表未果，进行广播,同时记录A在端口1

交换表1

MAC地址	端口
A	1

交换表2

MAC地址	端口

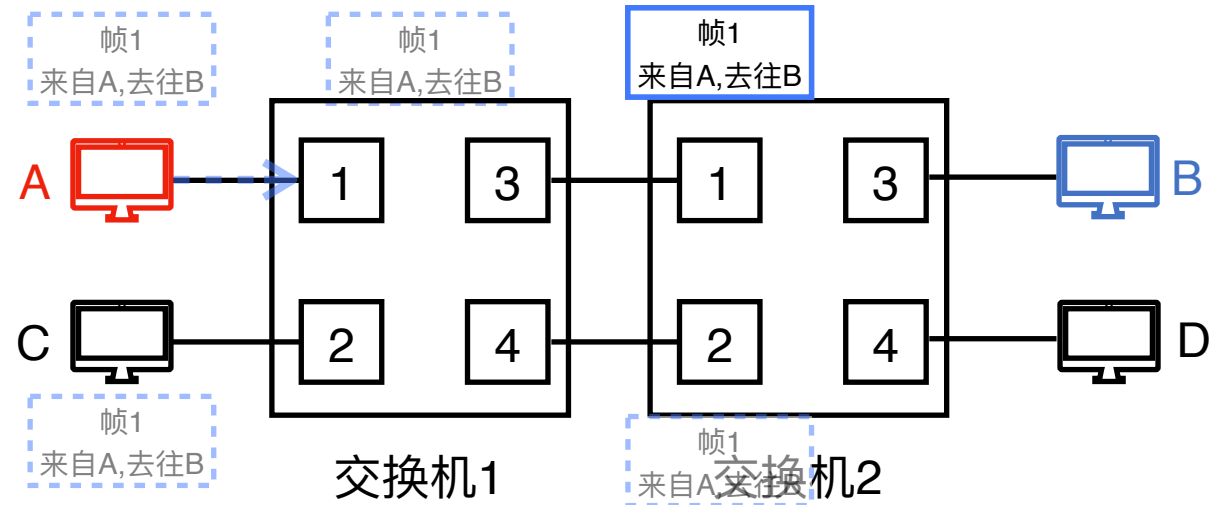
交换机1收到帧后, 查表未果, 进行广播,同时记录A在端口1

此时来到交换机2的帧，会因为交换机2中无B表项，又被广播发回A，从而无限循环

MAC地址	端口
A	1

MAC地址	端口

兜圈问题



A向B发送一帧

交换机1收到帧后，查表未果，进行广播,同时记录A在端口1

此时来到交换机2的帧，会因为交换机2中无B表项，又被广播发回A，从而无限循环

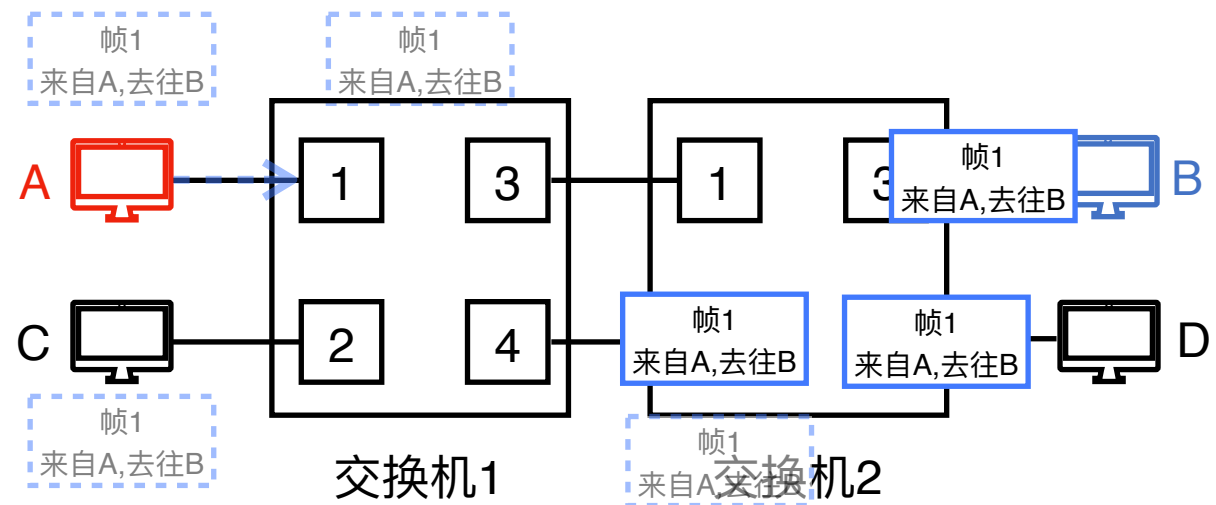
交换表1

MAC地址	端口
A	1

交换表2

MAC地址	端口

兜圈问题



A向B发送一帧

交换机1收到帧后，查表未果，进行广播,同时记录A在端口1

此时来到交换机2的帧，会因为交换机2中无B表项，又被广播发回A，从而无限循环

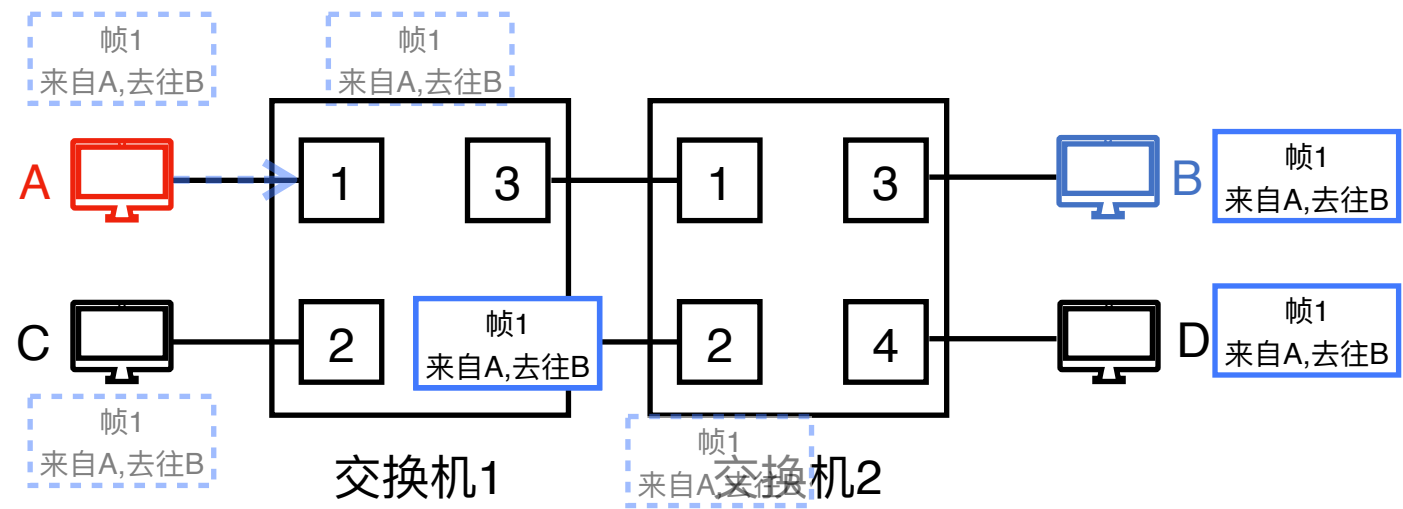
交换表1

MAC地址	端口
A	1

交换表2

MAC地址	端口

兜圈问题



A向B发送一帧

交换机1收到帧后，查表未果，进行广播,同时记录A在端口1

此时来到交换机2的帧，会因为交换机2中无B表项，又被广播发回A，从而无限循环

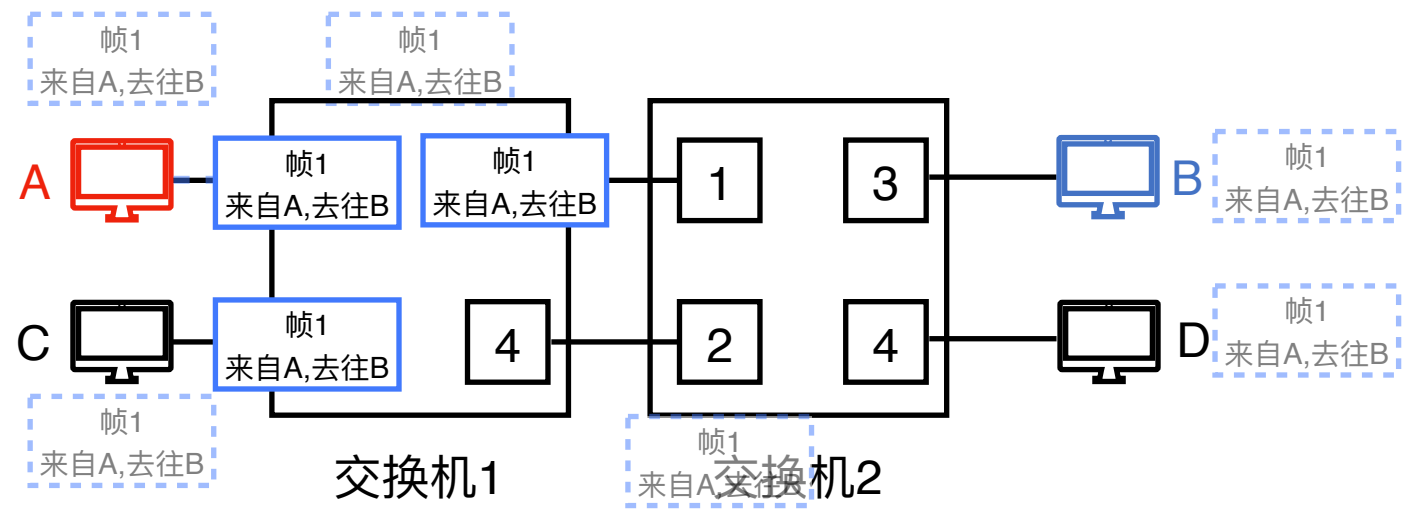
交换表1

MAC地址	端口
A	1

交换表2

MAC地址	端口

兜圈问题



A向B发送一帧

交换机1收到帧后，查表未果，进行广播,同时记录A在端口1

此时来到交换机2的帧，会因为交换机2中无B表项，又被广播发回A，从而无限循环

交换表1

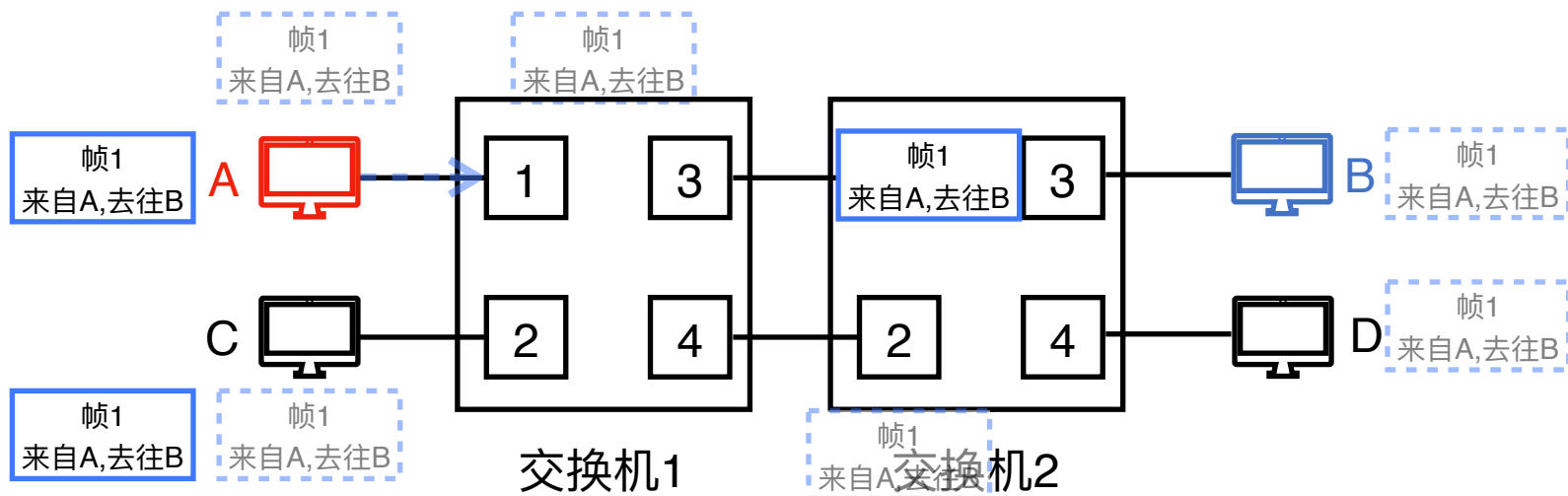
MAC地址	端口
A	1

交换表2

MAC地址	端口



兜圈问题

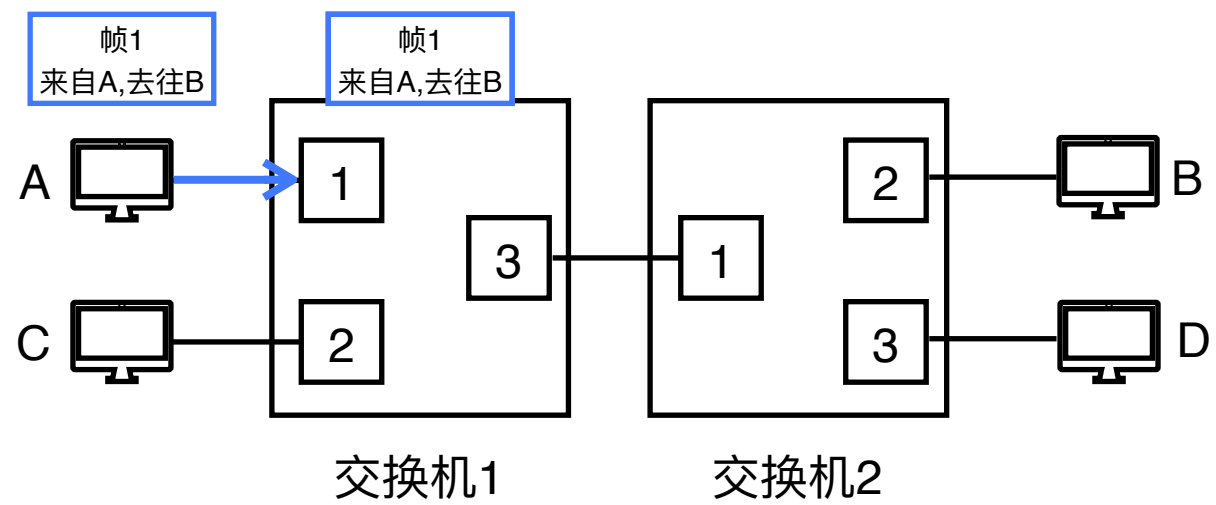


A向B发送一帧

交换机1收到帧后，查表未果，进行广播，同时记录A在端口1

此时来到交换机2的帧，会因为交换机2中无B表项，又被广播发回A，从而无限循环

对这种情况，我们会使用**生成树协议STP**来从逻辑上切断某些链路来避免兜圈
但是具体实现并非重点，了解即可



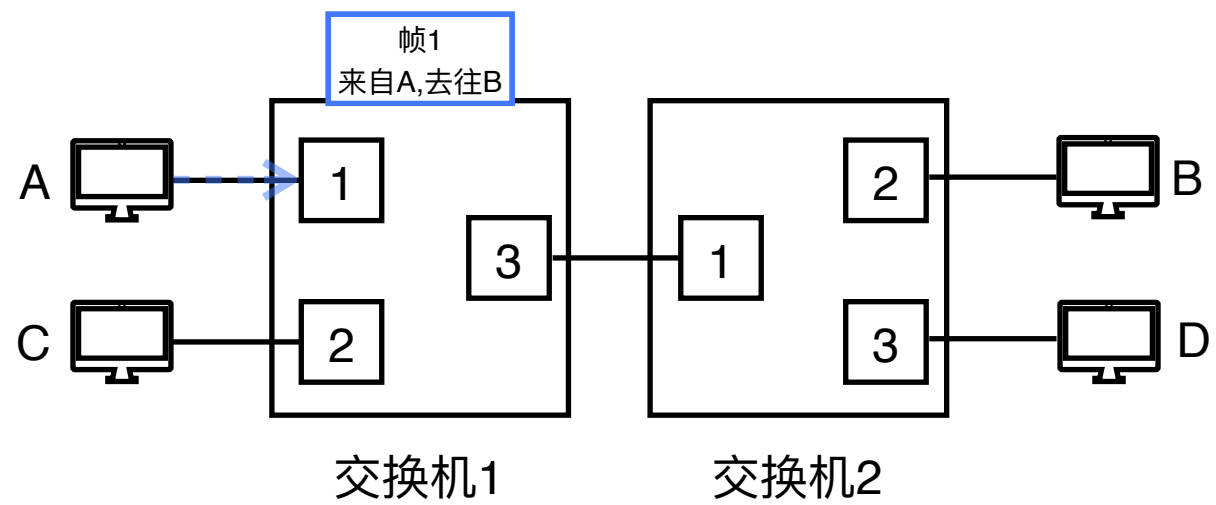
A向B发送一帧

交换表1

MAC地址	端口
A	1

交换表2

MAC地址	端口



A向B发送一帧

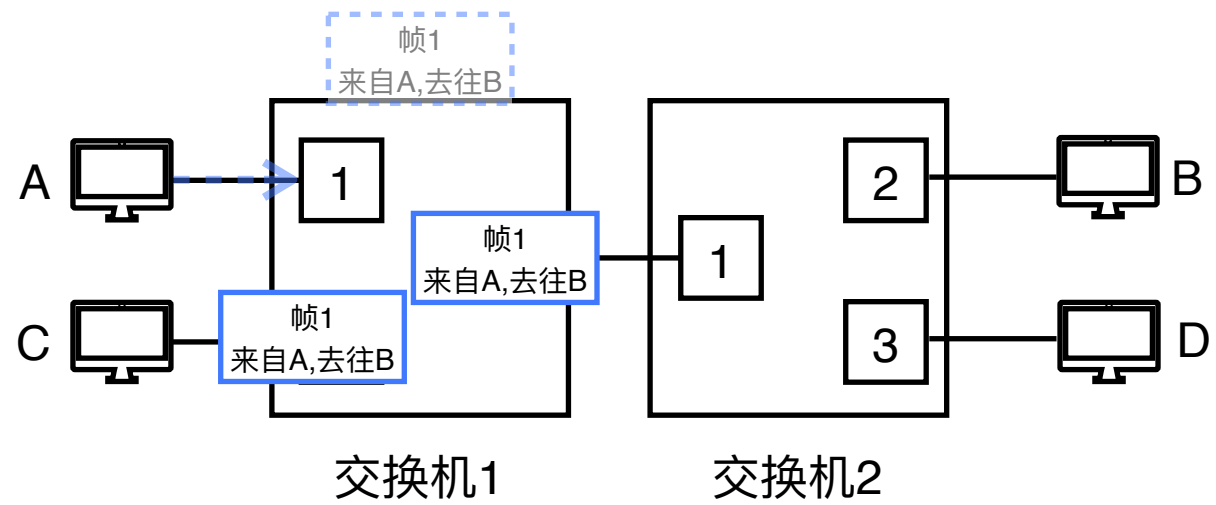
交换表1对A的mac地址进行记录后广播该帧

交换表1

MAC地址	端口
A	1

交换表2

MAC地址	端口



A向B发送一帧

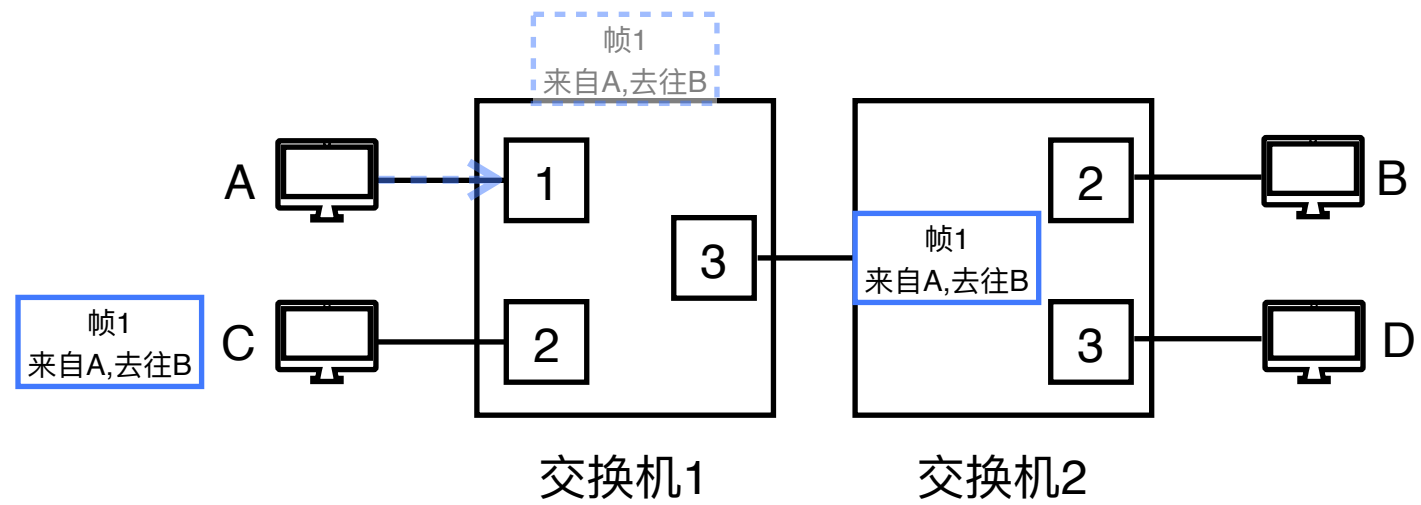
交换表1对A的mac地址进行记录后广播该帧

交换表1

MAC地址	端口
A	1

交换表2

MAC地址	端口



A向B发送一帧

交换表1对A的mac地址进行记录后广播该帧

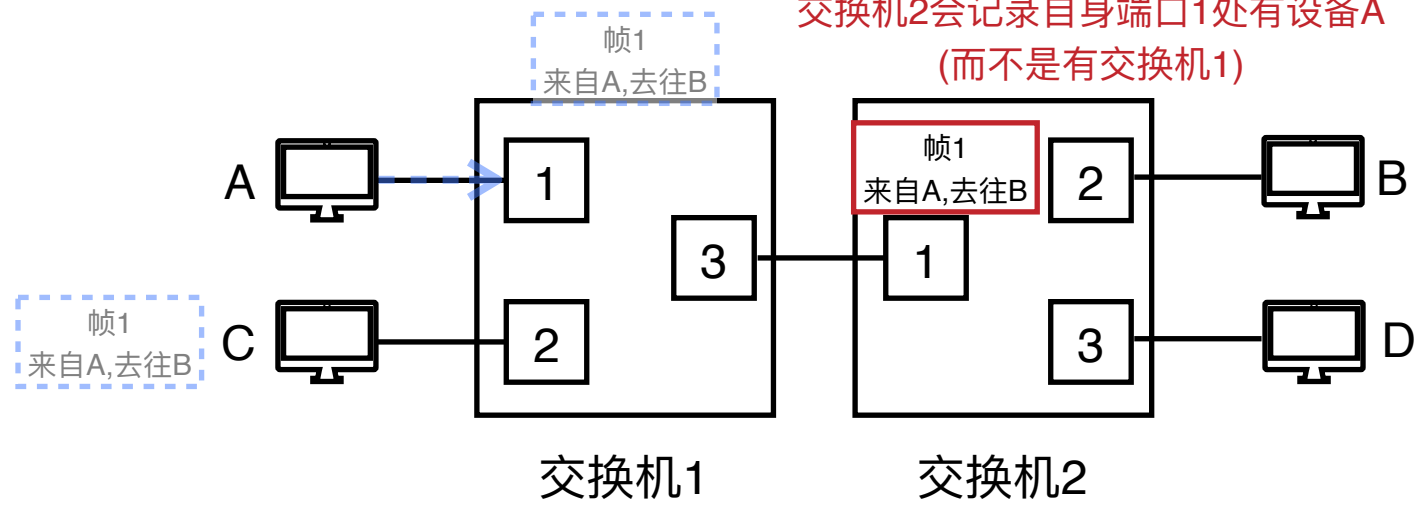
交换表1

MAC地址	端口
A	1

交换表2

MAC地址	端口

题目不说明的情况下默认交换机无mac地址
此时对于从交换机1来的帧1
交换机2会记录自身端口1处有设备A
(而不是有交换机1)



A向B发送一帧
交换表1对A的mac地址进行记录后广播该帧

交换表1

MAC地址	端口
A	1

交换表2

MAC地址	端口
A	1